

Lab: Philosophy of Science Through Active Inference

Learning Goal: Apply the rigorous formalisms of Active Inference to analyze the mechanics of theory change, formalize parsimony (Occam's razor), adjudicate demarcation criteria, and systematically address the problem of induction.

Part 1: Theory Change & Formal Paradigm Shifts

Exercise: Choose a major historical scientific revolution (e.g., Ptolemaic Geocentrism → Copernican Heliocentrism, Phlogiston theory → Oxygen theory, Newtonian Mechanics → General Relativity). Analyze the deep structure of this revolution by mapping it explicitly onto Active Inference mechanics:

Historical Phase	Philosophy of Science Concept	Active Inference Formalization	Application to your Case Study
Normal Science	Puzzle-solving within a paradigm	Continuous parameter learning (updating the $\mathbf{A}$ matrix)	
Anomaly	Persistent, unexplained observations	Chronically high sensory prediction errors ($\mathbf{S}^e$)	
Crisis	Loss of faith in the paradigm	Rising expected free energy ($\mathbf{G}$), precipitous drop in model evidence ($\ln P(\mathbf{y} \mathbf{m})$)	
Revolution	Adoption of a new paradigm	Bayesian model selection (structural update via Bayesian Model Reduction)	
Resolution	Return to normal science	Parameter learning within the newly selected, high-evidence model	

Write a 400-word critical analysis of your chosen case: Does the mathematical concept of Bayesian model selection genuinely capture the *sociological* and *political* dimensions of a Kuhnian paradigm shift, or does it overly intellectualize the messy history of science?

Write your response here

Part 2: Occam's Razor and Formal Parsimony

Learning Goal: Apply the mathematical necessity of the Complexity-Accuracy trade-off ($F \approx \text{Complexity} - \text{Accuracy}$) to adjudicate between competing models.

Exercise: Consider two competing generative models attempting to explain the exact same sensory dataset:

- **Model A (Simple):** 3 free parameters, Accuracy (log-likelihood) = 8.0, Complexity penalty = 2.0
- **Model B (Complex):** 30 free parameters, Accuracy (log-likelihood) = 8.5, Complexity penalty = 9.0
- **Calculation:** Compute the approximate variational free energy for each model.
- **Adjudication:** Which model does Occam's Razor mathematically favor, and why?
- **The Complexity Boundary:** At exactly what level of Accuracy would Model B scientifically justify its massive 30-parameter complexity? Show the math.
- **Machine Learning Parallel:** Explain in detail how this fundamental trade-off perfectly mirrors the concepts of "overfitting" and "regularization" (e.g., L1/L2 penalties) in contemporary deep learning architectures.

Write your response here

Part 3: The Demarcation Problem

Learning Goal: Apply the Active Inference framework to solve Karl Popper's classic demarcation criterion (distinguishing rigorous science from unfalsifiable pseudoscience).

Exercise: Analyze the following claims. Classify each as a scientific hypothesis, a pseudoscientific dogma, or an axiomatic prior using Active Inference criteria. Specifically, analyze the *precision* assigned to the prior belief versus the *precision* assigned to contradictory sensory prediction errors.

Claim	Classification	Precision Dynamics (Priors vs. Sensory PE)
"The Earth orbits the Sun in an elliptical path."		
"Your astrological birth chart dictates your personality."		
"Homeopathic remedies work via the 'memory' of water."		
"Evolution proceeds via natural selection."		
"A global, omnipotent conspiracy is secretly hiding the truth of flat earth."		

Write a 250-word synthesis: How does "dogmatic belief updating" (where infinite precision is assigned to top-down priors, instantly explaining away all bottom-up evidence) structurally define a conspiracy theory within a belief network?

Write your response here

Part 4: Induction and Evolutionary Priors

Exercise: David Hume's classic problem of induction asks: *By what rational justification can we expect the future to resemble the past?*

1. **The Problem:** State Hume's problem precisely in the language of probability theory.
2. **The Active Inference Solution:** Explain Active Inference's pragmatic, evolutionary response to induction. (Hint: The agent doesn't need a logical guarantee; it simply relies on phylogenetic priors forged by natural selection to minimize surprise).
3. **Philosophical Satisfaction:** Is this evolutionary response philosophically satisfying? Does it actually solve Hume's logical problem, or does it merely explain *why* human brains are forced to make inductive leaps to survive?
4. **Prior Selection:** How does this evolutionary history ultimately solve the infinite regress of prior selection in Bayesian epistemology?

Write your response here

Part 5: Critical Meta-Reflection

In 350 words, provide a conclusive reflection on the epistemology of the Free Energy

Principle: Does Active Inference provide a genuinely novel, mathematically unified account of the scientific method, or is it fundamentally a sophisticated post-hoc "curve-fitting" exercise that forcefully retrofits complex philosophy of science debates into a rigid thermodynamic metaphor?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept Analyzed
1	Historical formalization	Kuhnian Theory Change / Bayesian Model Selection
2	Quantitative reasoning	Occam's Razor / The Complexity Penalty
3	Epistemological mapping	The Demarcation Problem / Dogmatic Precision
4	Philosophical argumentation	The Problem of Induction / Phylogenetic Priors
5	Critical reflection	The Limits of Mathematical Formalization